

Percent Concentration - Practice Questions

Mass Over Mass Questions:

1. What is the % W/W if 8.0 g copper is added to enough zinc to produce 100 g of an alloy?
2. What is the % W/W of copper in an alloy when 10 kg of Cu is mixed with 250 kg of Zn?
3. Calculate the mass of solute in a 13% salt solution if the mass of the solution is 350 grams.

Volume over Volume Questions:

4. If you make a solution by adding 75 mL of ethanol to a container with 300 mL of water, what's the percent by volume of ethanol in the solution?
5. What is the percent by volume of isopropyl alcohol in a solution that contains 24 mL of isopropyl alcohol in 1.1 L of water.
6. Calculate the volume of the solute in a 20% NaCl solution if the volume of the solution is 350mL

Mass over volume Questions:

7. If 250 g of sodium chloride is present in 3 L of solution, what is the percentage weight by volume of sodium chloride?
8. Suppose you have 70g of sodium chloride salt (NaCl) in 250mL of water. Express this as a (W/V)% solution?
9. What is the volume of the solvent of a 25% solution where the mass of the solute is 12 g?

Ppm and ppb Questions:

10. What mass of nickel is in a 2.4 Kg sample of propanol if the concentration is 20 ppb?

11. What is the concentration of chloride in a solution made with 0.808 grams of CaCl_2 and 250.0 ml of water.? Give your answer in ppm.

Molarity Questions:

12. How many moles of Na_2CO_3 are there in 10.0 L of 2.00 M solution?

13. What is the molarity of a 0.654 L of solution with 0.444 mol of CoCl_2 in it?

14. What is the molarity of a 100.0 mL solution with 7.0×10^{-3} mol of I_2 in it?

Answers

1. 8% W/W
2. 3.8% W/W
3. 45.5 g
4. 20% V/V
5. 2.1% V/V
6. 70 mL
7. 8.3% W/V
8. 28% W/V
9. 48 mL
10. 4.8×10^{-11} kg
11. 2070 ppm
12. 20 moles
13. 0.679 M
14. 0.070 M